École Polytechnique Fédérale de Lausanne

Farm-Bike

Simultaneous Engineering Project

ME-314 Coursework

ME-314 Project Team

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Chapter 1

Preamble

A simultaneous engineering project at EPFL under the *Durabilité* unit, which aims to create a modular power source for various farm machinery utilizing human work as the source of power. The development of this machine was the subject of a simultaneous engineering project, course code ME-314, and is the work of a whole semester. The project describes every step from design to machining and testing.

Figure 1.1: Render of the machine.

The following README contains information that may overlap with other documentation contained in the repository but is not a replacement for those documents.

Chapter 2

Specification

2.1 Functionality

- Provide a way of converting human power into useful shaft work.
- Have easily adjustable transmission ratio.
- Allow for quick coupling of multiple machines.

2.2 Work Environment

- Resist an outdoor work environment subject to a central European climate.
- Resist storage in wet conditions.

2.3 Durability

- Be built wherever possible out of standard parts to ensure long term parts availability.
- Machined components to only use conventional machining to provide possibility of field repair.
- Use of quality components to ensure a long life cycle.
- Easy maintenance procedures.

Chapter 3

Design

The design of the machine was driven by the specification outlined above and the will to ensure the project can be replicated or further developed in the future. The design choices are driven by the will to create a machine that has a long service life, quality parts and material were chosen over low-cost alternatives.

3.1 Frame

The frame is made out of Item modular aluminium extrusions, specifically the profile 8 series. This made the construction of the frame simple and required only off-the-shelf parts. The Item profile 8 can be substituted by any equivalent 40 mm × 40 mm T-slot aluminium extrusion. The choice of aluminium extrusion was made to ensure availability of materials in any geographical region as well as ensure repairability, the ease of manufacture was of course a welcome bonus.

3.2 Drivetrain

A transmission based on bicycle components is the optimal off-the-shelf solution to our problem. It offers robustness, ease of maintenance and worldwide availability at a low cost. The following components in the system required no modification from their retail configuration:

- Cassette.
- Derailleur & Shifter.
- Handlebars & Stem.
- Crank set & Bottom Bracket.
- Seat post & Saddle.

The machine is designed to be used as a modular power source for farming implements, many of which are hand-cranked and operate around 60 rpm. The bicycle drivetrain and pulley dimensions were selected so that the operator can adjust the rotational speed without stopping the machine. With the chosen cassette, chainring and pulley diameters, and for an assumed pedalling cadence of 70 rpm, the machine output can vary between approximately 47 rpm and 137 rpm. If another farming implement requires a different speed range, the driven pulley diameter can be changed accordingly.

3.3 Gear Ratios and Speed Calculations

The speed calculations are present on the spreadsheet included in the GitHub repository under `Documentation/FarmBikeSpeedCalc.xlsx`. The assumed rider input is a cadence of 70 rpm and a mechanical power of 250 W. The crank torque is therefore calculated as

$$T_{\text{crank}} = \frac{P\,60}{n\,2\pi} = 34.10\ \text{N m}.$$

The high-speed case uses the smallest cassette sprocket, while the low-speed case uses the largest cassette sprocket. The final pulley stage uses a 71 mm drive pulley and a 112 mm driven pulley.

Table 3.1: Gear ratio, speed and torque calculations for the Farm-Bike drivetrain.

Parameter	High-speed setting	Low-speed setting
Chainring teeth	34	34
Cassette sprocket teeth	11	32
Bicycle drivetrain ratio $i_{\text{chain}} = N_{\text{chainring}}/N_{\text{cassette}}$	3.09	1.06
Hub speed at 70 rpm cadence [rpm]	216.36	74.38
Hub torque for 250 W input [N m]	11.03	32.09
Drive pulley diameter [mm]	71	71
Driven pulley diameter [mm]	112	112
Pulley ratio $i_{\text{pulley}} = D_{\text{drive}}/D_{\text{driven}}$	0.63	0.63
Overall speed ratio $i_{\text{tot}} = i_{\text{chain}} i_{\text{pulley}}$	1.96	0.67
Machine output speed [rpm]	137.16	47.15
Machine output torque [N m]	17.40	50.63

For the currently selected driven pulley diameter of 112 mm, this range covers the operating speed of the polenta mill and corn sheller used during testing. In the spreadsheet you can also see a centre-to-centre distance of 1000 mm, which gives an estimated belt length of approximately 2288 mm for the 71 mm and 112 mm pulley pair.

The subsequent paragraphs describe the design of each of the machine-specific parts. The parts were designed to conform with bicycle industry standards such that commissioning and maintenance could be carried out by anyone familiar with common bicycles.

3.3.1 Bottom Bracket Shell

The bottom bracket shell attaches the standard BSA bicycle bottom bracket to the frame. The part is a weldment of a round bottom bracket shell recovered from a donor frame and a 40 mm × 30 mm rectangular steel tube. The part requires far less material resources than an equivalent machined from a billet and does not require specialty tooling to cut the BSA thread.

3.3.2 Hub

The hub forms a central part of the transmission of the machine, it mounts the cassette and the pulley thus transmitting all of the power generated. A flywheel is also mounted on the axle to smooth out the power delivery.

3.3.3 Pillow Block

This part houses the main bearings and attaches to the frame. It is a crucial part that maintains the alignment between the two bearings and ensures their long life and smooth operation, for this reason it is machined out of a 6061 aluminium billet. Due to unavailability of appropriate stock the pillow block has been split into two pieces and pinned together. It features clearance holes for two M8 mounting bolts, a central bore for the axle to pass through and two counterbores for the 6005-2Z bearings. These deep groove ball bearings have been selected to ensure a long life span as well being easy to source. Low profile ball bearings often found in bicycles have specifically been discarded as their lower profile makes them more prone to premature failure. A recess has been added at the mounting between the pillow block and the aluminium profile to prevent twisting during usage.

3.3.4 Axle

The large bearings allow us to use an aluminium axle without risking the keyway becoming fragile. The axle features 25 mm bearing seats and a key way to transmit torque to the flywheel and pulley. In order to mount the cassette onto the axle while avoiding complex machining operations a donor freehub body was bonded onto the axle. The axle was of course designed around the specific freehub body available thus the design would need to be modified in accordance with available parts.

3.3.5 Derailleur Hanger

To emulate the derailleur mounting of a conventional bicycle without having to cut a fine pitch thread a derailleur hanger from a donor frame was utilized. In order to correctly position it with respect to the axle and cassette a mount was designed out of 3 mm 5005 aluminium sheet. The mount exploits the advantages of bent sheet metal to achieve a robust yet lightweight solution.

Chapter 4

Manufacturing

This section outlines the manufacturing processes of each of the parts and provides photos of the components. It assumes the reader is comfortable with conventional machining processes and general fabrication and thus omits the details of most trivial operations. Some parts in the assembly are overlooked in this section as their manufacturing features no operations of interest.

4.1 Bottom Bracket Shell

This is the only weldment on the machine, consisting of two parts with thin sections. The donor frame was chosen for its steel construction and BSA threaded bottom bracket. The shell was cut out and turned down to a constant diameter. During the turning operations a close fitting aluminium plug was used on the inside of the threads to provide tailstock support. The hole for the shell in the rectangular tubing was roughed out with a hole saw on a mill and fitted by hand with a file. The two parts were subsequently TIG welded and the thread was chased. TIG welding was used because of the thin material and in order to preserve as much of the BSA thread inside the shell.

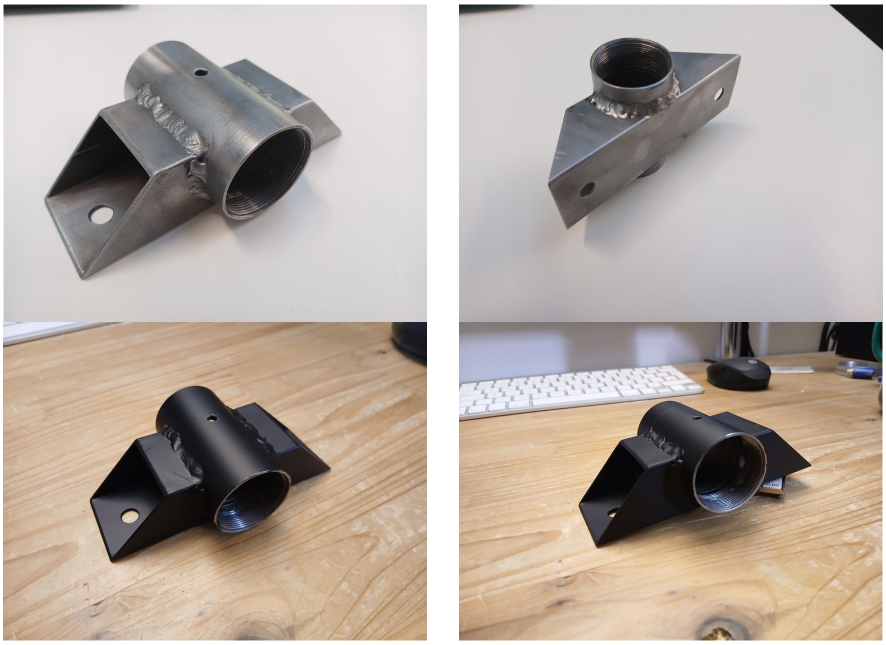


Figure 4.1: Bottom bracket shell manufacturing steps.

After welding, the oxides were removed and the part was finished with automotive anti-rust primer.

4.2 Axle

The machining operations did not deviate from standard practice. Loctite 603 retaining compound was used to chemically bond the freehub body to the axle. Chemical bonding was chosen to remove the need for pressing in the axle thus eliminating the risk of fracture on the hardened freehub body.

4.3 Pillow Block

Due to limitations in the available stock material the pillow block was machined from two separate pieces of stock.

The operations were carried out in the following order:

- Squaring the stock to size.
- Drilling and reaming of the 4 mm pin bores.
- Centre drilling of the main bore.
- Two parts are pressed together.
- Milling locating slot.
- Drilling of two M8 clearance holes.
- Boring half of the through bore in a lathe with centre drill hole as reference.
- Boring of bearing seat.
- Part flipped and boring operations repeated.
- Deburring and chamfering.

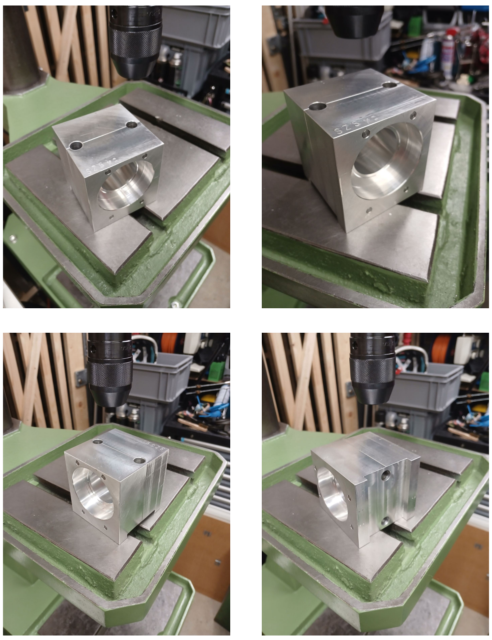


Figure 4.2: Pillow block manufacturing steps.

Following machining, deburring and finishing was carried out and the bearings were fitted.

4.4 Fly Wheel

In order to avoid procuring costly stock material the fly wheel is cut out of 5 mm S235 structural steel and the pieces are pinned together. The use of a fibre laser allows for significant time saving but all of the parts can be machined using conventional processes. The parts were pressed together with an arbor press and a coat of anti-rust primer protects against corrosion.

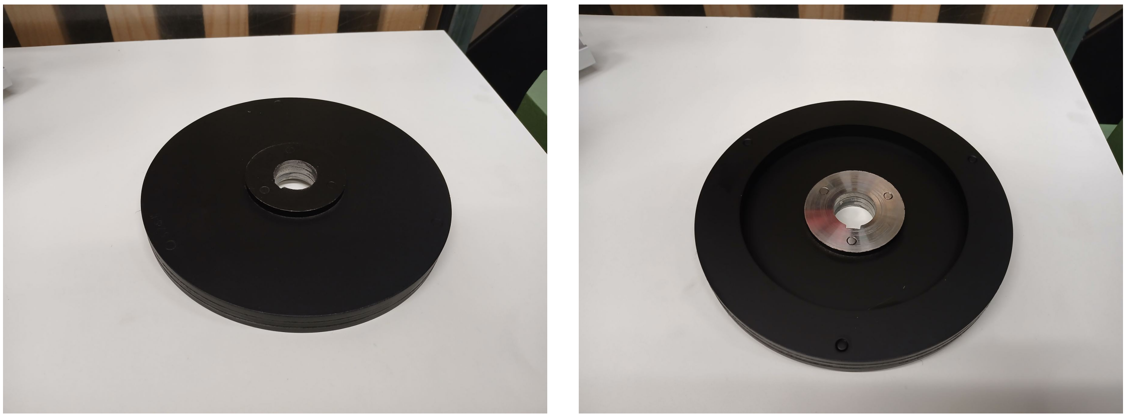


Figure 4.3: Fly wheel manufacturing and assembly.

4.5 Hub

Precautions taken during assembly of the hub:

- Parts cleaned and lubricated.
- Light preload applied to the bearings.
- Taper lock pulley and bushing torqued to manufacturer specification.

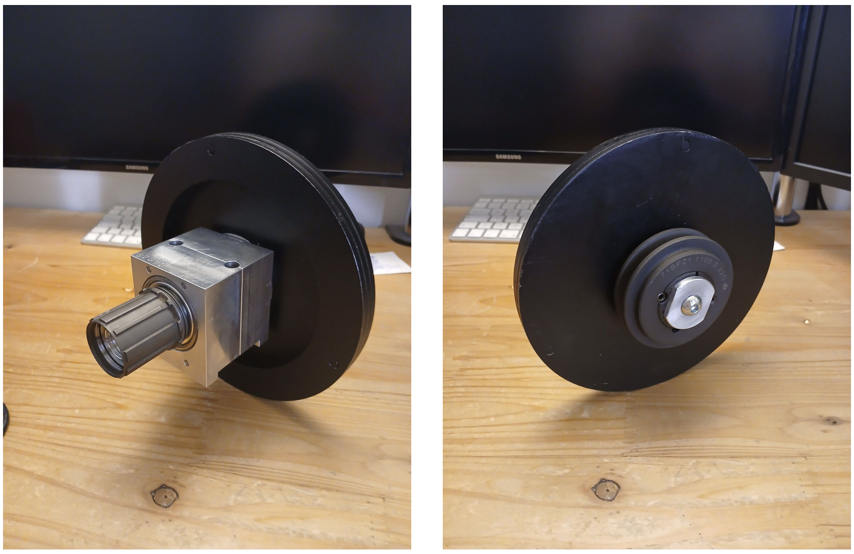


Figure 4.4: Hub assembly.

4.6 Derailleur Mount

Just like the fly wheel the derailleur mount was cut with the use of a fibre laser. Due to its non-trivial geometry the sheet metal part was bent in a bench vice with the help of parallel jaw pliers. The existing derailleur hanger was hand fitted during the finishing process. As most of the part geometries are non-mating surfaces the part could be replicated with hand tools. The geometry can be modified to suit the manufacturing methods available.

Figure 4.5: Derailleur mount manufacturing and fitting.

Chapter 5

Procurement Guide

Normalized industrial parts for this project can be commonly bought new from well established suppliers. The parts for the prototype were sourced from Norelem, but equivalent parts can be sourced from:

- Misumi.
- RS Components.
- McMaster-Carr.
- JLCMC.

To reduce the ecological impact of the prototype second hand or recovered components were utilized in areas that would not impact the longevity, safety or operation of the machine.

Item aluminium profiles and relevant hardware were recovered from decommissioned machinery. Equivalents can be sourced from the suppliers above or Item24 directly. Wherever possible material off-cuts or scrapped parts were used as stock material.

Bicycle components for the prototype were second hand and sourced from *Point Vélo* located on the EPFL campus.

Chapter 6

User Guide

6.1 Use Instructions

Before operating the machine, ensure that it is placed on a stable and level surface. The frame should not rock or move under load, and all fasteners securing the aluminium profiles, pillow block, bottom bracket shell, seat post and handlebar assembly should be checked for tightness. The pulley, flywheel and cassette should be visually inspected to ensure that they are correctly seated and that no foreign object is interfering with the drivetrain.

The user should adjust the saddle height and handlebar position to obtain a comfortable pedalling position. As with a conventional bicycle, the saddle height should allow the leg to remain slightly bent at the bottom of the pedal stroke. The drivetrain can then be shifted using the standard bicycle shifter in order to select the desired transmission ratio. The rear derailleur can be shifted using the standard handlebar shifter, while the front chainring selection has to be carried out manually. The user should therefore stop pedalling, wait for the drivetrain to come to rest, and move the chain by hand only when the machine is fully stationary. This should be done by getting off the bike, adjusting the chain to the desired chainring setting and gently cranking the pedals by hand such that the chain jumps into its desired place. The user can afterwards get back onto the bike and pedal normally. Lower gears should be used when starting the machine or when driving a high-load implement, while higher gears can be used once the system is already rotating smoothly or when a higher output speed is required.

To start the machine, the user should begin pedalling progressively and avoid sudden impacts on the pedals. The flywheel will gradually store kinetic energy and smooth out the power delivery. Once the machine is rotating steadily, the selected gear ratio can be adjusted depending on the required output speed and the resistance of the coupled machine. Gear changes should preferably be carried out while pedalling with moderate force, in the same way as on a standard bicycle, in order to avoid excessive stress on the chain and derailleur.

Before coupling the Farm-Bike to an external machine, the user must ensure that the driven machine is compatible with the output speed, torque and rotation direction of the system. The coupling should be securely mounted and aligned with the pulley or output shaft to avoid vibrations, belt slipping or premature wear. During operation, hands, clothing and loose objects must be kept away from the chain, cassette, pulley, flywheel and any rotating element.

To stop the machine, the user should simply reduce pedalling effort progressively and allow the drivetrain and flywheel to slow down naturally. The user should not attempt to stop the flywheel, pulley or chain by hand. After use, the machine should be inspected for loose fasteners, abnormal noise, chain misalignment or signs of wear. If the machine has been used outdoors or stored in humid conditions, exposed steel parts should be dried and checked for corrosion.

6.2 Maintenance Guide

Time intervals for preventative maintenance are guidelines only. Maintenance intervals should be adjusted according to the conditions of use and storage.

- Before each use: inspect fasteners and drivetrain.
- Every 7–10 hours: clean and lubricate the chain.
- Every 30–50 hours: degrease and lubricate the chain.
- Every 100 hours: inspect the drivetrain for wear.
- Every 150 hours: disassemble and inspect the hub and crankset.

Parts deemed inoperable during inspection are to be replaced. For a more detailed explanation of fitting, adjusting and maintaining bicycle-specific components, refer to the General Operations Manual, *DM-GN0001*, published by Shimano.

6.3 Troubleshooting Guide

Table 6.1: Common operating issues and likely causes.

Issue	Likely cause	Corrective action
Unusual noise	Loose fastener	Verify and tighten fasteners.
Chain skips under power	Worn drivetrain components	Replace the chain. If the issue persists, replace the cassette and chainrings.
Chain jumps between gears	Shifting not adjusted	Adjust the shift cable.
Chain falls off the top or bottom of the cassette	Derailleur limits out of adjustment	Adjust the derailleur limits.
Excessive friction in drivetrain under no load	Debris stuck or tangled in the drivetrain	Inspect the drivetrain and remove debris.

Chapter 7

Testing

7.1 Initial Testing: 14/05/2026 (Week 12)

The first functional test was carried out directly at the Bassenges farm. The Farm-Bike prototype was brought to the farm and connected to two existing machines: the polenta grinder and the corn sheller. Pulleys were installed on both machines, after which the bike was coupled to the system and tested. The first test was successful, as the Farm-Bike was able to drive the machines through the belt transmission. The whole Bassenges team was present during the test.

However, once the system was operating, an important ergonomic issue was identified by Timothée, the farmer responsible for following the project at the Bassenges farm. In the initial configuration, the bike was not facing the machines. This meant that a single operator could not comfortably pedal the bike and load the machines at the same time. This feedback highlighted an important practical limitation of the first assembly.

After considering the problem, we realized that the modular aluminium profile frame made it possible to rearrange the bike without manufacturing new parts. Several components were repositioned so that the pulley attachment faced the opposite direction and was located at the front of the bike. This reassembly took approximately an hour and a half, but it allowed the Farm-Bike to be adapted directly to the farmer's request. Due to the lack of equipment at the farm, we were not able to reduce the length of the chain, which is planned for the following week.

Initial testing confirmed the value of using modular aluminium profiles for the frame. The ability to modify the geometry of the machine on site was a major advantage during the testing phase. If the frame had been fully welded or made as a fixed steel structure, this last-minute correction would not have been possible without significant rework.

The following images show the Farm-Bike before and after the reassembly:

Initial configuration *Modified configuration*

Figure 7.1: Farm-Bike configuration before and after the first testing session at Bassenges farm.

Appendix A

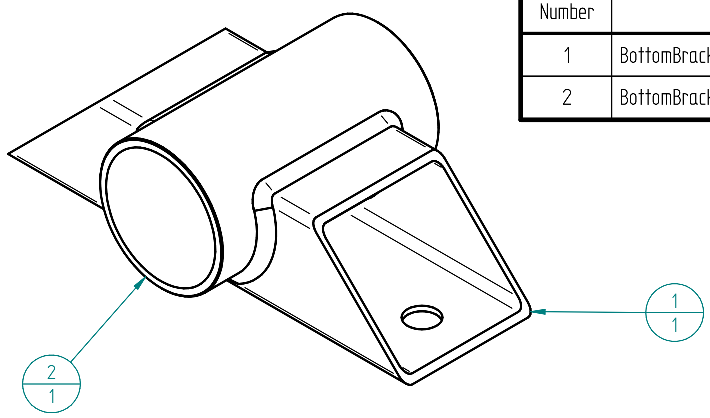
Technical Drawing Annex

This annex gathers the PDF technical drawings stored in the `Drafts` section of the GitHub repository.

A.1 BBMount

REVISION HISTORY			
REV	DESCRIPTION	DATE	APPROVED

Item Number	File Name (no extension)	Author	Quantity
1	BottomBracketMount	Franek	1
2	BottomBracketShell	Franek	1



Bottom Bracket Shell can be recovered from donor frame

	NAME	DATE
DRAWN	Franek	04/19/26
CHECKED		
ENG APPR		
MGR APPR		

UNLESS OTHERWISE SPECIFIED
DIMENSIONS ARE IN MILLIMETERS
ANGLES ±X.X°
2 PL ±X.XX 3 PL ±X.XXX

Solid Edge

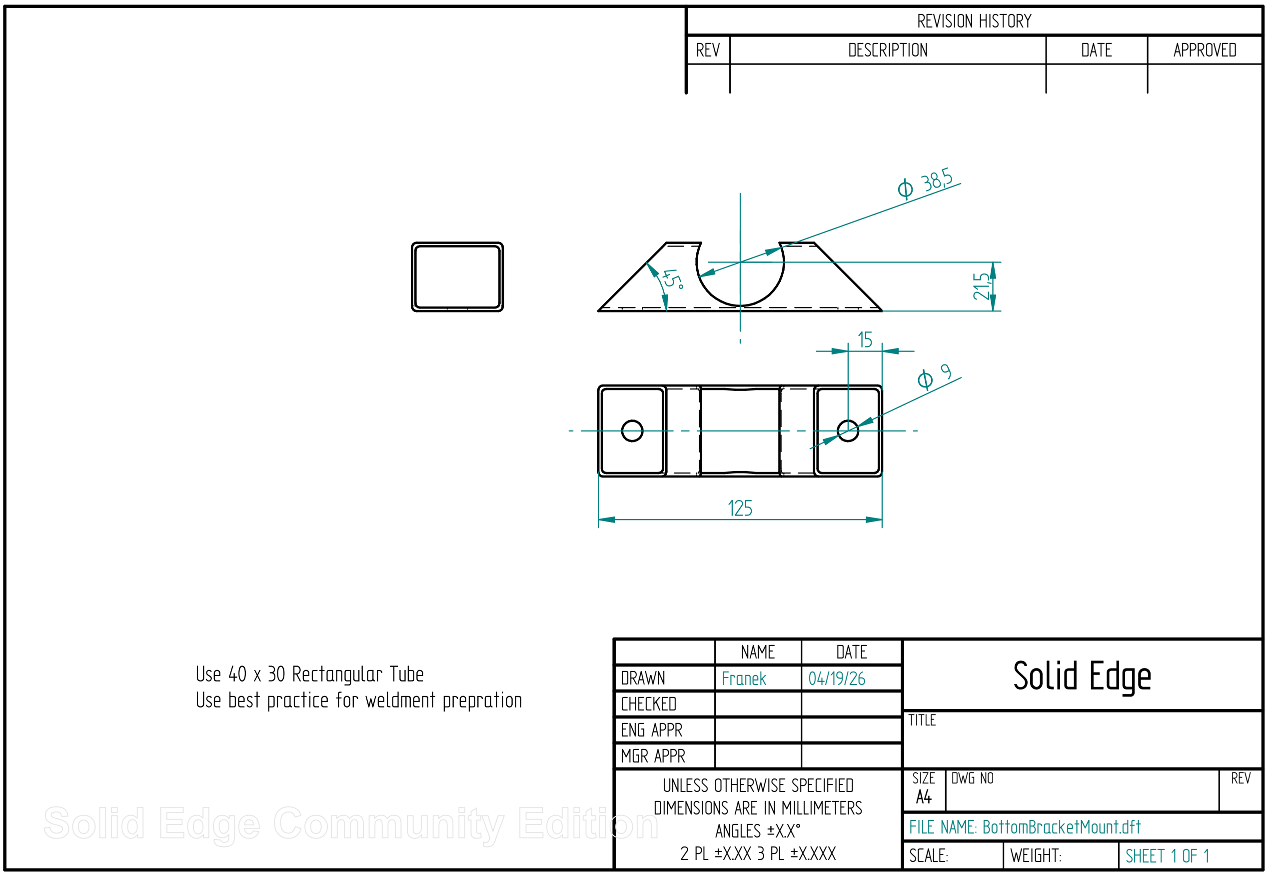
TITLE

SIZE	DWG NO	REV
A4		

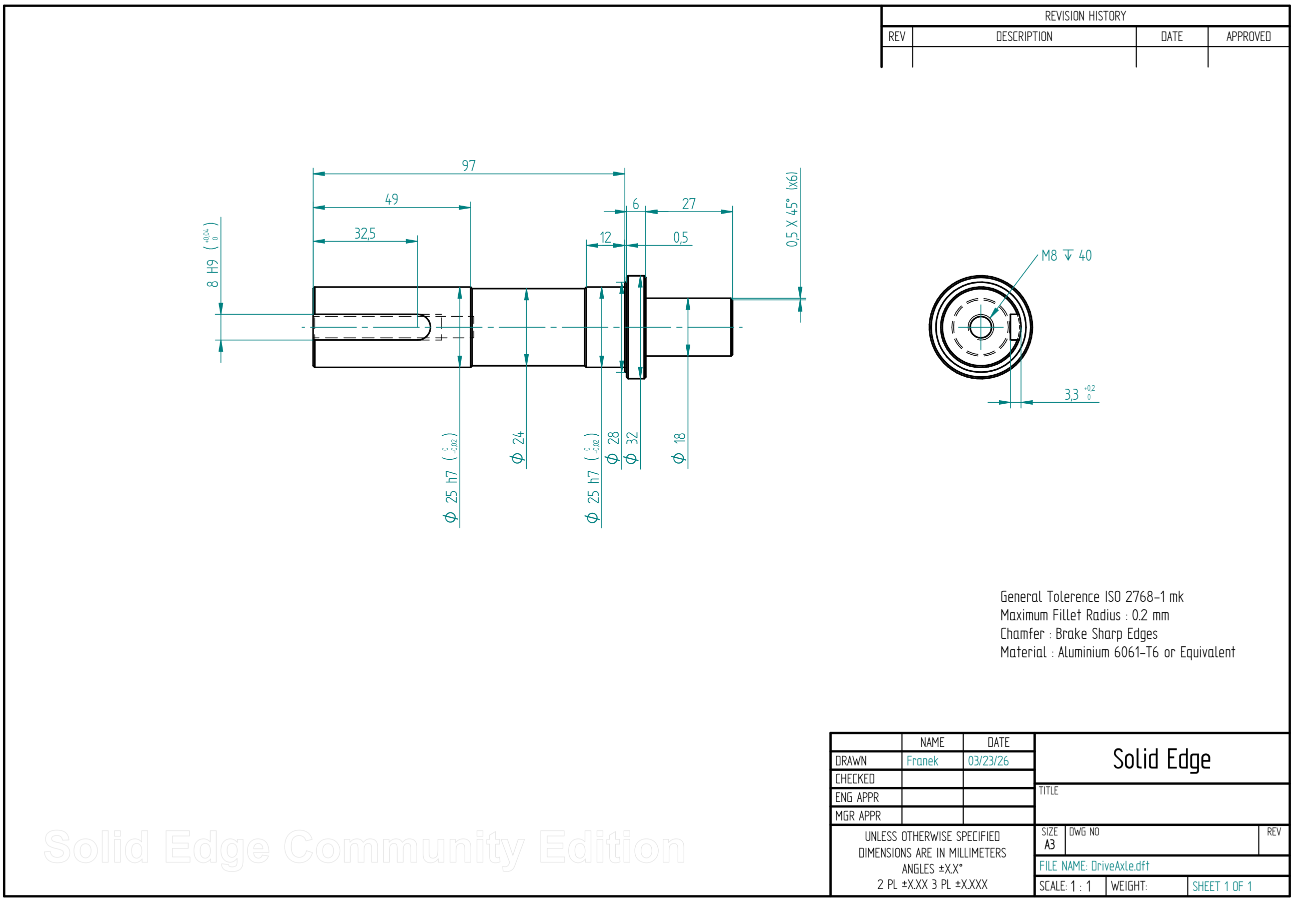
FILE NAME: BBMount.dft

SCALE:	WEIGHT:	SHEET 1 OF 1

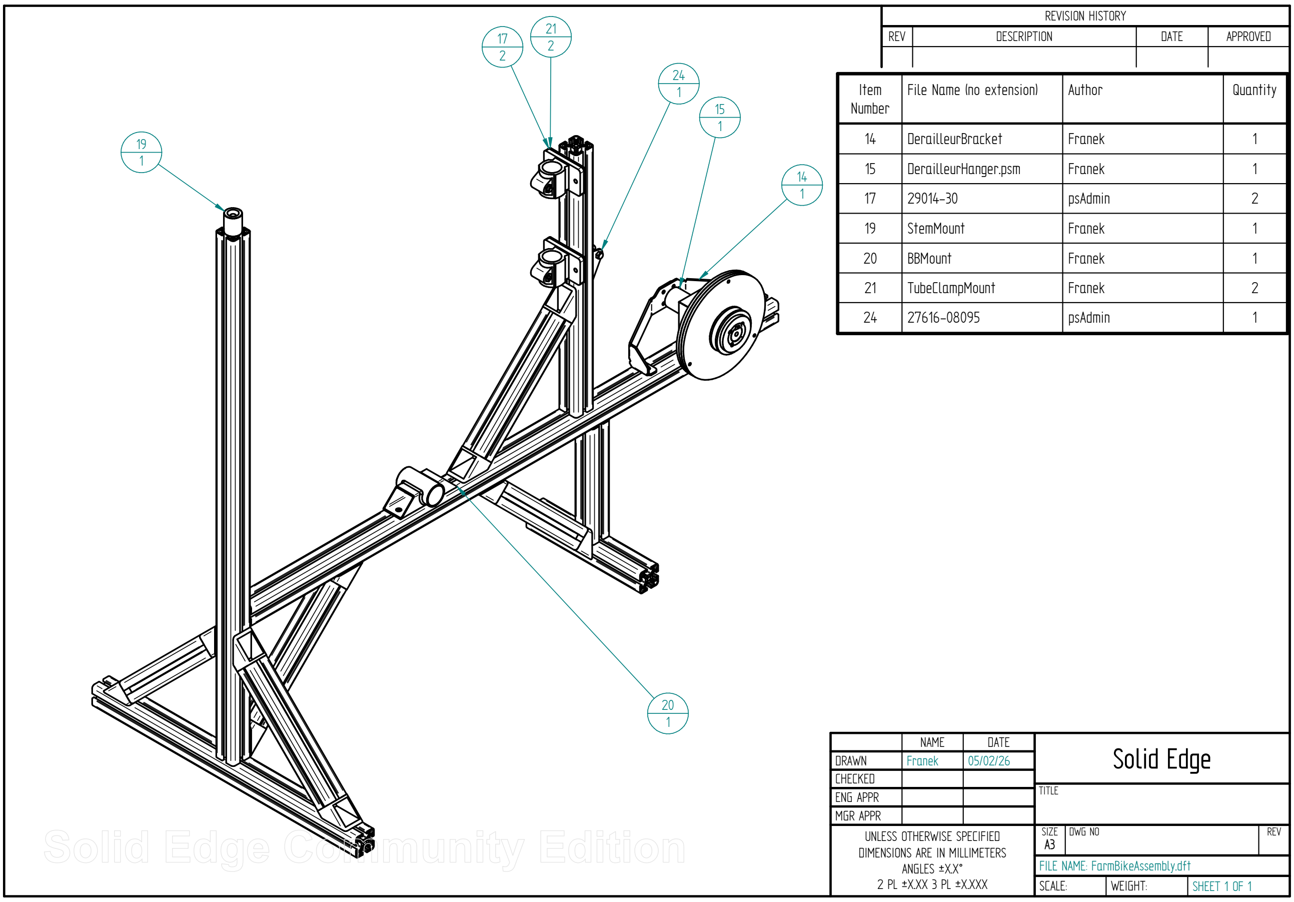
A.2 Bottom Bracket Mount



A.3 Drive Axle

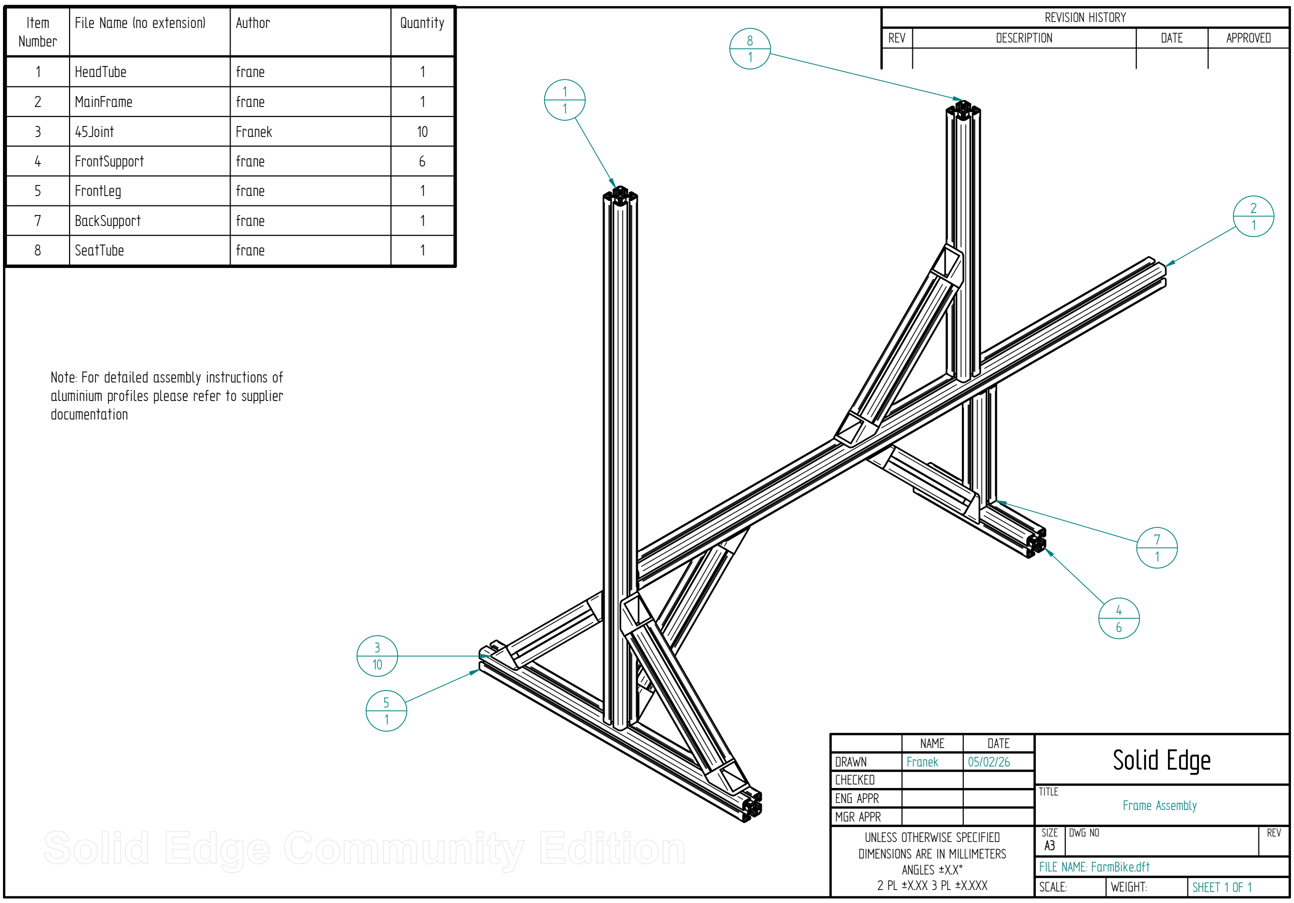


A.4 Farm Bike Assembly



Item Number	File Name (no extension)	Author	Quantity
14	DerailleurBracket	Franek	1
15	DerailleurHanger.psm	Franek	1
17	29014-30	psAdmin	2
19	StemMount	Franek	1
20	BBMount	Franek	1
21	TubeClampMount	Franek	2
24	27616-08095	psAdmin	1

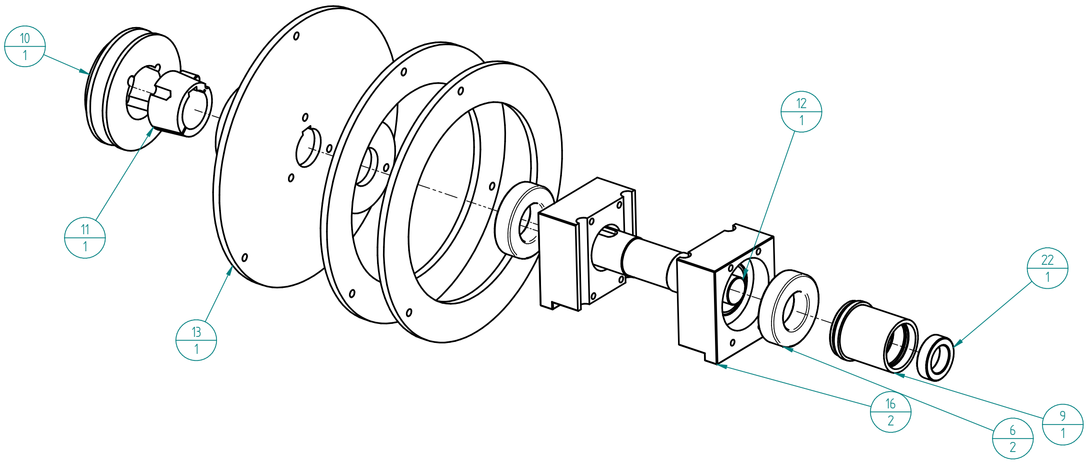
A.5 Frame Assembly



Item Number	File Name (no extension)	Author	Quantity
1	HeadTube	frane	1
2	MainFrame	frane	1
3	45Joint	Franek	10
4	FrontSupport	frane	6
5	FrontLeg	frane	1
7	BackSupport	frane	1
8	SeatTube	frane	1

A.6 Hub Exploded

REVISION HISTORY			
REV	DESCRIPTION	DATE	APPROVED



Item Number	File Name (no extension)	Author	Quantity
6	6005-2Z	Franek	2
9	FreeHubBody	Franek	1
10	22070-12711	psAdmin	1
11	23200-0382522	psAdmin	1
12	DriveAxle	Franek	1
13	FlyWheelAssy	Franek	1
16	PillowBlock	Franek	2
22	DummyBearing	Franek	1

Assembly Note: Parts #9, #12, #22 are to be bonded with retaining compound Loctite 603 or equivalent

Note: Dowel pins and drive key omitted for drawing clarity

	NAME	DATE
DRAWN	Franek	05/02/26
CHECKED		
ENG APPR		
MGR APPR		

UNLESS OTHERWISE SPECIFIED
DIMENSIONS ARE IN MILLIMETERS
ANGLES ±X.X°
2 PL ±X.XX 3 PL ±X.XXX

Solid Edge

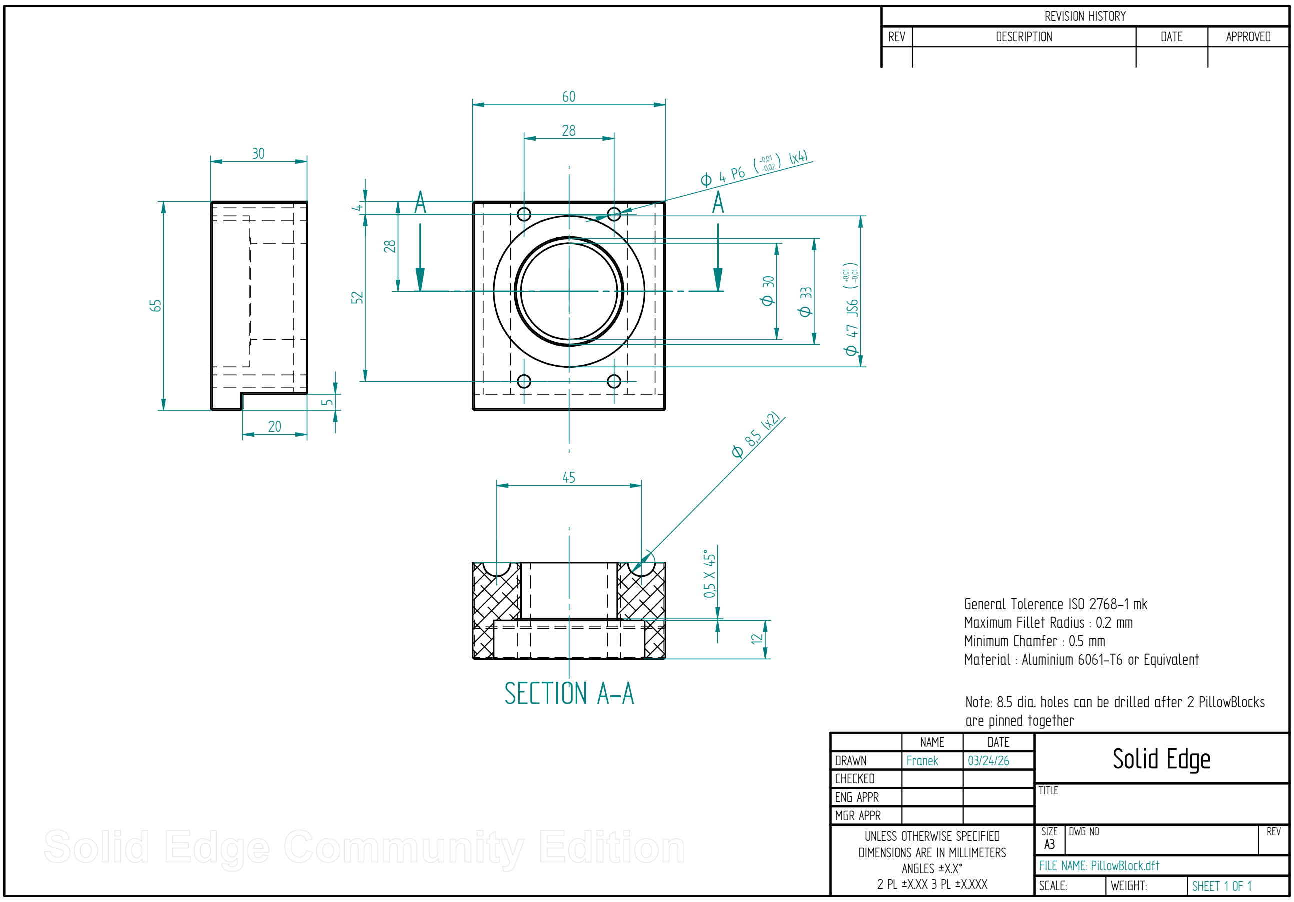
TITLE
Hub Assembly

SIZE A3 | DWG NO | REV

FILE NAME: HubExploded.dft

SCALE: | WEIGHT: | SHEET 1 OF 1

A.7 Pillow Block



A.8 Stem Mount